# Cross-Stream and Cross-Channel Attention Networks for Surgical Skill Classification in Open Surgery From Hand Kinematics.

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## Abstract

Despite the extensive research on skill assessment in minimally invasive surgery, applications in open surgery (OS) remain limited. Twenty trainees performed three OS tasks-knot tying (KT), continuous suturing (CS), and interrupted suturing (IS)-yielding 201 trials. Various deep learning models based on LSTM and Transformer were evaluated for binary skill classification using hand kinematics. The proposed architecture integrates cross-stream and cross-channel attention mechanisms to capture inter-hand, intra-hand, and stream-level motion interactions. Agreement with an independent reviewer was also assessed. The inter-hand model achieved the best performance across all tasks and outperformed the reviewer in multiple metrics (e.g., Accuracy: 0.88 vs. 0.86 KT; 0.84 vs. 0.83 CS; 0.81 vs. 0.68 IS). High-skill recognition was better for KT, whereas low-skill recognition was better for the more demanding tasks, CS and IS. Our study highlights the importance of capturing hand kinematic relationships as key indicators of surgical performance.

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